A Universal Square-Root-of-Six Bound for Kissing Numbers

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Abstract

Let τ_n denote the kissing number in Euclidean dimension n. We give an elementary proof that

$$\tau_n \leq (\sqrt{6})^n \qquad (n \geq 1).$$

The constant $\sqrt{6}$ is optimal among bounds of the form $\tau_n \leq \alpha^n$ required to hold in every dimension, since $\tau_2 = 6$. The argument combines the elementary spherical-cap packing bound with a two-step recurrence for cap-volume fractions. We also state precisely the scope of an accompanying Lean 4/Mathlib certificate.

AI provenance and verification status. The proof and the first draft of this manuscript were generated by OpenAI's GPT-5.6 Pro from a prompt supplied by the author. The accompanying Lean 4/Mathlib project checks the numerical and inductive core of the argument, conditional on explicitly stated geometric and analytic inputs. It is not yet an end-to-end formalization of the kissing-number theorem; the exact scope is recorded in Section 3. The present version has not undergone independent human mathematical review.

1 Statement of the bound

For $n \geq 1$, let τ_n be the largest number of nonoverlapping unit balls in $\mathbb{R}^n$ that can simultaneously touch a fixed unit ball. Equivalently, τ_n is the largest cardinality of a subset of the unit sphere S^{n-1} whose pairwise angular distances are at least $\pi/3$.

The following answers the universal-base question that motivated this note.

Theorem 1. *For every integer $n \geq 1$,*

$$\tau_n \leq (\sqrt{6})^n.$$

Corollary 2. *The least real number $\alpha > 0$ such that*

$$\tau_n \leq \alpha^n \qquad \text{for every } n \geq 1$$

is $\sqrt{6}$. Equivalently,

$$\sup_{n\geq 1} \tau_n^{1/n} = \sqrt{6}.$$

2 Proof

For $n \geq 2$, define

$$c_n := \frac{\displaystyle\int_0^{\pi/6} \sin^{n-2} t\, dt}{\displaystyle\int_0^{\pi} \sin^{n-2} t\, dt}. \tag{1}$$

This is the normalized surface measure of a spherical cap of angular radius $\pi/6$ in S^{n-1}. Indeed, polar coordinates about the center of the cap give the surface element as a constant multiple of $\sin^{n-2} t\, dt$, and the constant cancels in the ratio (1).

If the centers of two such caps have angular distance at least $\pi/3$, then their interiors are disjoint. The boundaries have surface measure zero, so every kissing configuration gives

$$\tau_n c_n \le 1. \tag{2}$$

We next obtain a uniform lower bound for c_n.

Lemma 3. *For every $n \ge 2$,*

$$c_{n+2} > \frac{c_n}{6}.$$

Proof. Put $m = n - 2$ and set

$$I_m := \int_0^{\pi/6} \sin^m t\, dt, \qquad J_m := \int_0^{\pi} \sin^m t\, dt,$$

so that $c_n = I_m/J_m$. Integration by parts gives

$$I_{m+2} = \frac{m+1}{m+2} I_m - \frac{\sqrt{3}}{(m+2)2^{m+2}}. \tag{3}$$

$$J_{m+2} = \frac{m+1}{m+2} J_m. \tag{4}$$

Consequently,

$$c_{n+2} = c_n - \frac{\sqrt{3}}{(m+1)2^{m+2} J_m}. \tag{5}$$

It remains to bound I_m from below. With the substitution $u = \sin t$,

$$I_m = \int_0^{1/2} \frac{u^m}{\sqrt{1-u^2}}\, du.$$

For $0 \le u \le 1/2$,

$$\frac{1}{\sqrt{1-u^2}} \ge 1 + \frac{u^2}{2}. \tag{6}$$

To see this, both sides are nonnegative and

$$(1-u^2)\left(1 + \frac{u^2}{2}\right)^2 = 1 - \frac{3u^4}{4} - \frac{u^6}{4} \le 1.$$

It follows that

$$\begin{aligned} I_m &\ge \int_0^{1/2} u^m \left(1 + \frac{u^2}{2}\right) du \\ &= \frac{1}{2^m(m+1)} \left(\frac{1}{2} + \frac{m+1}{16(m+3)}\right) \\ &\ge \frac{25}{48\, 2^m (m+1)}. \end{aligned}$$

Since $\sqrt{3} < 26/15$, we have

$$\frac{3\sqrt{3}}{10} < \frac{13}{25} < \frac{25}{48},$$

and hence

$$I_m > \frac{3\sqrt{3}}{10(m+1)2^m}. \tag{7}$$

Using $J_m > 0$, inequality (7) yields

$$\frac{\sqrt{3}}{(m+1)2^{m+2} J_m} < \frac{5}{6} \frac{I_m}{J_m} = \frac{5}{6} c_n.$$

Substitution into (5) proves $c_{n+2} > c_n/6$. $\square$

Proof of Theorem 1. First suppose that n is even. Since $c_2 = 1/6$, Lemma 3 gives

$$c_n \geq 6^{-n/2}.$$

Together with (2), this implies

$$\tau_n \leq \frac{1}{c_n} \leq 6^{n/2} = (\sqrt{6})^n.$$

For odd $n \geq 5$, direct integration gives

$$c_5 = \frac{1}{2} - \frac{9\sqrt{3}}{32}.$$

The estimate $\sqrt{3} < 26/15$ implies $c_5 > 1/80$, whereas

$$(\sqrt{6})^5 = 36\sqrt{6} > 80$$

because $\sqrt{6} > 20/9$. Thus $c_5 > 6^{-5/2}$, and another application of Lemma 3 gives

$$c_n > 6^{-n/2} \qquad \text{for every odd } n \geq 5.$$

Equation (2) again yields the desired bound.

It remains to consider $n = 1$ and $n = 3$. In dimension one, $\tau_1 = 2 < \sqrt{6}$. In dimension three,

$$c_3 = \frac{2 - \sqrt{3}}{4} > \frac{1}{15},$$

where the last inequality follows from $\sqrt{3} < 26/15$. Hence (2) gives $\tau_3 < 15$, so the integrality of τ_3 gives $\tau_3 \leq 14$. Finally,

$$14 < 6\sqrt{6} = (\sqrt{6})^3,$$

because $\sqrt{6} > 7/3$. This completes the proof. □

Proof of Corollary 2. Theorem 1 shows that $\alpha = \sqrt{6}$ works. Conversely, $\tau_2 = 6$: six equally spaced points on S^1 form a kissing configuration, while pairwise angular separation at least $\pi/3$ allows at most six points around the circle. Therefore every universal base α satisfies

$$6 = \tau_2 \leq \alpha^2,$$

and hence $\alpha \geq \sqrt{6}$. □

Remark 4. The argument is deliberately elementary and is not intended to compete asymptotically with the best high-dimensional spherical-code bounds. Its point is that $\sqrt{6}$ is the optimal constant in a dimension-uniform estimate having the particularly simple form $\tau_n \leq \alpha^n$ with no additional prefactor.

3 Scope of the Lean certificate

The accompanying repository is

`https://github.com/sebastienpalcoux/sqrt6-kissing-bound`.

It pins Lean 4 and Mathlib, runs a proof-hole scan and an axiom audit, and checks the following parts of the proof:

- the numerical inequalities involving $\sqrt{3}$ and $\sqrt{6}$;

- the base estimate at $n = 5$;
- the two-step induction from $c_{n+2} \geq c_n/6$;
- cancellation of a positive cap fraction in $Nc_n \leq 1$;
- the exceptional numerical argument in dimension three;
- optimality of $\sqrt{6}$ from the inequality $6 \leq \alpha^2$.

The present certificate does not yet formalize the definition of kissing configurations, the spherical-cap packing argument and cap-area formula, the integration-by-parts identities (3)–(4), the lower estimate (7), or the low-dimensional identities $\tau_1 = 2$ and $\tau_2 = 6$. These appear in the formal source as explicit hypotheses or structured inputs, rather than as hidden or project-defined axioms. The repository file `FORMALIZATION_SCOPE.md` gives a declaration-by-declaration account.

Acknowledgment

This note was prompted by the MathOverflow question "Upper bound of the kissing number in n dimensions" and by Henry Cohn's answer explaining how effective spherical-code bounds bear on the problem.

References

[1] S. Palcoux, *Upper bound of the kissing number in n dimensions*, MathOverflow question 282644 (2017), `https://mathoverflow.net/questions/282644/`.

[2] H. Cohn, *Answer to "Upper bound of the kissing number in n dimensions"*, MathOverflow answer 282685 (2017), `https://mathoverflow.net/a/282685/`.

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